

## LUMINA vs. Trans

| Title                 |        | FP   | FN | TP  | TN   | Total | FP Rate | FN Rate | Specificity | Sensitivity |
|-----------------------|--------|------|----|-----|------|-------|---------|---------|-------------|-------------|
| Sommer<br>(Version 2) | LUMINA | 870  | 0  | 11  | 4579 | 5460  | 0.16    | 0       | 0.84        | 1           |
|                       | Trans  | 776  | 22 | 98  | 3181 | 4077  | 0.196   | 0.183   | 0.804       | 0.817       |
| Kiesswetter           | LUMINA | 328  | 0  | 16  | 6480 | 6824  | 0.048   | 0       | 0.952       | 1           |
|                       | Trans  | 2828 | 11 | 66  | 3573 | 6478  | 0.4418  | 0.1429  | 0.5582      | 0.8571      |
| Sbidian               | LUMINA | 1941 | 0  | 167 | 3272 | 5380  | 0.372   | 0       | 0.628       | 1           |
|                       | Trans  | 3215 | 27 | 747 | 1115 | 5104  | 0.74249 | 0.00349 | 0.25751     | 0.96511     |

|        |        |     |    |     |      |      |          |          |          |             |
|--------|--------|-----|----|-----|------|------|----------|----------|----------|-------------|
| Sommer | LUMINA | 513 | 0  | 22  | 1983 | 2518 | 0.206    | 0        | 0.794    | 1           |
|        | Trans  | 187 | 44 | 189 | 252  | 672  | 0.425968 | 0.188841 | 0.574032 | 0.811158798 |

- Sommerl, Dobrescu A, Ledinger D, et al. Outpatient treatment of confirmed COVID-19: a living, rapid review for the American College of Physicians. Ann Intern Med. 2023;176:92-104. [PMID: 36442056]doi:10.7326/M22-2202 24.
- Sommer I, Ledinger D, Thaler K, et al. Outpatient treatment of confirmed COVID-19: a living, rapid evidence review for the American College of Physicians (Version 2). Ann Intern Med. 2023;176:1377-1385. [PMID: 37722115] doi:10.7326/M23-1626 25.
- Kiesswetter E, Stadelmaier J, Petropoulou M, et al. Effects of dairy intake on markers of cardiometabolic health in adults: a systematic review with network meta-analysis. Adv Nutr. 2023;14:438 450. [PMID: 36914032]doi:10.1016/j.advnut.2023.03.004
- Sbidian E, Chaimani A, Guelimi R, et al. Systemic pharmacological treatments for chronic plaque psoriasis: a network meta-analysis. Cochrane Database Syst Rev. 2023;7:CD011535.

## All 15 SRMAs LUMINA Results

| Number of included articles in the original paper | Title       | FP   | FN | TP | TN    | Total | FP Rate | FN Rate | Sensitivity | Specificity |
|---------------------------------------------------|-------------|------|----|----|-------|-------|---------|---------|-------------|-------------|
| 32                                                | Reference 1 | 1437 | 1  | 28 | 11495 | 12961 | 0.111   | 0.034   | 0.966       | 0.888       |
| 12                                                | Reference 2 | 264  | 0  | 12 | 8250  | 8526  | 0.031   | 0       | 1           | 0.967       |
| 113                                               | Reference 3 | 1163 | 2  | 89 | 4539  | 5793  | 0.204   | 0.022   | 0.978       | 0.796       |
| 45                                                | Reference 4 | 205  | 0  | 31 | 11420 | 11656 | 0.018   | 0       | 1           | 0.982       |

|    |              |     |   |    |      |      |       |      |      |       |
|----|--------------|-----|---|----|------|------|-------|------|------|-------|
| 9  | Reference 5  | 234 | 0 | 8  | 2183 | 2425 | 0.097 | 0    | 1    | 0.903 |
| 16 | Reference 6  | 167 | 1 | 13 | 1105 | 1286 | 0.13  | 0.07 | 0.93 | 0.87  |
| 15 | Reference 7  | 228 | 0 | 15 | 1656 | 1899 | 0.12  | 0    | 1    | 0.88  |
| 13 | Reference 8  | 538 | 0 | 10 | 2896 | 3444 | 0.157 | 0    | 1    | 0.843 |
| 14 | Reference 9  | 63  | 0 | 11 | 509  | 583  | 0.11  | 0    | 1    | 0.89  |
| 10 | Reference 10 | 94  | 0 | 8  | 2462 | 2564 | 0.037 | 0    | 1    | 0.963 |
| 10 | Reference 11 | 19  | 0 | 10 | 135  | 164  | 0.123 | 0    | 1    | 0.844 |

|     |                 |     |   |    |       |       |       |       |       |       |
|-----|-----------------|-----|---|----|-------|-------|-------|-------|-------|-------|
| 28  | Reference<br>12 | 277 | 1 | 23 | 7827  | 8128  | 0.034 | 0.04  | 0.96  | 0.966 |
| 27  | Reference<br>13 | 760 | 0 | 19 | 11159 | 11938 | 0.064 | 0     | 1     | 0.936 |
| 25  | Reference<br>14 | 65  | 1 | 12 | 250   | 328   | 0.187 | 0.077 | 0.923 | 0.794 |
| 111 | Reference<br>15 | 585 | 2 | 81 | 1160  | 1828  | 0.335 | 0.02  | 0.98  | 0.665 |

### Result Matrix:

|             | mean     | std      | min   | 25%    | 50%   | 75%    | max   | range |
|-------------|----------|----------|-------|--------|-------|--------|-------|-------|
| FP Rate     | 0.1172   | 0.082629 | 0.018 | 0.0505 | 0.111 | 0.1435 | 0.335 | 0.317 |
| FN Rate     | 0.017533 | 0.026535 | 0     | 0      | 0     | 0.028  | 0.077 | 0.077 |
| Sensitivity | 0.982467 | 0.026535 | 0.923 | 0.972  | 1     | 1      | 1     | 0.077 |

|             |          |          |          |          |          |          |          |          |
|-------------|----------|----------|----------|----------|----------|----------|----------|----------|
| Specificity | 0.879133 | 0.084311 | 0.665    | 0.8435   | 0.888    | 0.9495   | 0.982    | 0.317    |
| NPV         | 0.999516 | 0.001081 | 0.996016 | 0.999716 | 1        | 1        | 1        | 0.003984 |
| PPV         | 0.093381 | 0.083204 | 0.018248 | 0.038268 | 0.072222 | 0.126489 | 0.344828 | 0.326579 |

### Methodological Insights into ChatGPT's Screening Performance in Systematic Reviews

| Title       | FP   | FN | TP | TN    | Total | FP Rate | FN Rate | Sensitivity | Specificity |
|-------------|------|----|----|-------|-------|---------|---------|-------------|-------------|
| Reference 1 | 666  | 15 | 39 | 12241 | 12961 | 0.0516  | 0.2778  | 0.7222      | 0.9484      |
| Reference 2 | 78   | 5  | 18 | 8425  | 8526  | 0.0092  | 0.2174  | 0.783       | 0.9908      |
| Reference 3 | 2148 | 4  | 90 | 3551  | 5793  | 0.3769  | 0.0426  | 0.9574      | 0.6231      |
| Reference 4 | 224  | 3  | 42 | 11387 | 11656 | 0.0193  | 0.0667  | 0.9333      | 0.9807      |
| Reference 5 | 152  | 2  | 11 | 2259  | 2424  | 0.063   | 0.1538  | 0.8462      | 0.937       |
| Reference 6 | 81   | 4  | 22 | 1179  | 1286  | 0.0643  | 0.1538  | 0.8462      | 0.9357      |

|                 |     |    |    |       |       |        |        |        |        |
|-----------------|-----|----|----|-------|-------|--------|--------|--------|--------|
| Reference<br>7  | 227 | 3  | 26 | 1642  | 1898  | 0.1215 | 0.1034 | 0.8966 | 0.8785 |
| Reference<br>8  | 509 | 0  | 14 | 2921  | 3444  | 0.1484 | 0      | 1      | 0.8516 |
| Reference<br>9  | 101 | 1  | 14 | 467   | 583   | 0.1778 | 0.0667 | 0.9333 | 0.8222 |
| Reference<br>10 | 90  | 0  | 13 | 2461  | 2564  | 0.0353 | 0      | 1      | 0.9647 |
| Reference<br>11 | 31  | 5  | 15 | 113   | 164   | 0.2153 | 0.25   | 0.75   | 0.7847 |
| Reference<br>12 | 95  | 13 | 31 | 7989  | 8128  | 0.0118 | 0.2955 | 0.7055 | 0.9882 |
| Reference<br>13 | 139 | 10 | 13 | 11776 | 11938 | 0.0117 | 0.4348 | 0.5652 | 0.9883 |



|                 |     |    |     |      |      |        |        |        |        |
|-----------------|-----|----|-----|------|------|--------|--------|--------|--------|
| Reference<br>14 | 5   | 9  | 4   | 310  | 328  | 0.0159 | 0.6923 | 0.3077 | 0.9841 |
| Reference<br>15 | 619 | 10 | 132 | 1067 | 1828 | 0.3671 | 0.0704 | 0.9296 | 0.6329 |

### Result Matrix:

|             | mean   | std    | min    | 25%    | 50%    | 75%    | max    | range  |
|-------------|--------|--------|--------|--------|--------|--------|--------|--------|
| FP Rate     | 0.0903 | 0.0651 | 0.0092 | 0.0193 | 0.0516 | 0.1215 | 0.3769 | 0.3677 |
| FN Rate     | 0.1694 | 0.1825 | 0      | 0.0426 | 0.0667 | 0.1538 | 0.6923 | 0.6923 |
| Sensitivity | 0.8495 | 0.092  | 0.3077 | 0.75   | 0.8462 | 0.9333 | 1      | 0.6923 |
| Specificity | 0.9069 | 0.0705 | 0.6231 | 0.7847 | 0.9333 | 0.9647 | 0.9908 | 0.3677 |
| NPV         | 0.9972 | 0.0043 | 0.9802 | 0.9868 | 0.9991 | 0.9997 | 0.9999 | 0.0197 |
| PPV         | 0.1373 | 0.0922 | 0.0273 | 0.0851 | 0.1208 | 0.1875 | 0.4444 | 0.4171 |

### Evaluating the effectiveness of large language models in abstract screening: a comparative analysis

| Title       | FP  | FN | TP | TN    | Total | FP Rate     | FN Rate     | Sensitivity | Specificity |
|-------------|-----|----|----|-------|-------|-------------|-------------|-------------|-------------|
| Reference 1 | 420 | 19 | 35 | 12487 | 12961 | 0.032540482 | 0.351851852 | 0.648148148 | 0.967459518 |
| Reference 2 | 21  | 22 | 1  | 8482  | 8526  | 0.002469717 | 0.956521739 | 0.043478261 | 0.997530283 |
| Reference 3 | 0   | 94 | 0  | 5699  | 5793  | 0           | 1           | 0           | 1           |
| Reference 4 | 95  | 10 | 35 | 11516 | 11656 | 0.008181896 | 0.222222222 | 0.777777778 | 0.991818104 |
| Reference 5 | 17  | 4  | 9  | 2394  | 2424  | 0.007051016 | 0.307692308 | 0.692307692 | 0.992948984 |
| Reference 6 | 42  | 20 | 6  | 1218  | 1286  | 0.033333333 | 0.769230769 | 0.230769231 | 0.966666667 |

|                 |    |    |    |       |       |             |             |             |             |
|-----------------|----|----|----|-------|-------|-------------|-------------|-------------|-------------|
| Reference<br>7  | 74 | 14 | 15 | 1795  | 1898  | 0.039593365 | 0.482758621 | 0.517241379 | 0.960406635 |
| Reference<br>8  | 33 | 9  | 5  | 3397  | 3444  | 0.009620991 | 0.642857143 | 0.357142857 | 0.990379009 |
| Reference<br>9  | 3  | 11 | 4  | 565   | 583   | 0.00528169  | 0.733333333 | 0.266666667 | 0.99471831  |
| Reference<br>10 | 22 | 4  | 9  | 2529  | 2564  | 0.008624069 | 0.307692308 | 0.692307692 | 0.991375931 |
| Reference<br>11 | 1  | 18 | 2  | 143   | 164   | 0.006944444 | 0.9         | 0.1         | 0.993055556 |
| Reference<br>12 | 58 | 18 | 26 | 8026  | 8128  | 0.007174666 | 0.409090909 | 0.590909091 | 0.992825334 |
| Reference<br>13 | 29 | 22 | 1  | 11886 | 11938 | 0.002433907 | 0.956521739 | 0.043478261 | 0.997566093 |

|                 |     |    |    |      |      |             |             |             |             |
|-----------------|-----|----|----|------|------|-------------|-------------|-------------|-------------|
| Reference<br>14 | 0   | 13 | 0  | 315  | 328  | 0           | 1           | 0           | 1           |
| Reference<br>15 | 232 | 58 | 84 | 1454 | 1828 | 0.137603796 | 0.408450704 | 0.591549296 | 0.862396204 |

### Result Matrix:

|             | mean        | std         | min         | 25%         | 50%         | 75%         | max         | range       |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| FP Rate     | 0.020056892 | 0.034909545 | 0           | 0.003875703 | 0.007174666 | 0.021080737 | 0.137603796 | 0.137603796 |
| FN Rate     | 0.629881576 | 0.287990516 | 0.222222222 | 0.380151278 | 0.642857143 | 0.92826087  | 1           | 0.777777778 |
| Sensitivity | 0.370118424 | 0.287990516 | 0           | 0.07173913  | 0.357142857 | 0.619848722 | 0.777777778 | 0.777777778 |
| Specificity | 0.979943108 | 0.034909545 | 0.862396204 | 0.978919263 | 0.992825334 | 0.996124297 | 1           | 0.137603796 |
| NPV         | 0.982402551 | 0.02912549  | 0.888198758 | 0.982338149 | 0.997357604 | 0.998242223 | 0.999132396 | 0.110933639 |
| PPV         | 0.253844481 | 0.192957969 | 0.033333333 | 0.125       | 0.265822785 | 0.30952381  | 0.666666667 | 0.633333333 |

### Prompting is all you need: LLMs for systematic review screening

| Title       | FP  | FN | TP | TN    | Total | FP Rate     | FN Rate     | Sensitivity | Specificity |
|-------------|-----|----|----|-------|-------|-------------|-------------|-------------|-------------|
| Reference 1 | 857 | 13 | 41 | 12050 | 12961 | 0.066398079 | 0.240740741 | 0.759259259 | 0.933601921 |
| Reference 2 | 83  | 16 | 7  | 8420  | 8526  | 0.009761261 | 0.695652174 | 0.304347826 | 0.990238739 |
| Reference 3 | 544 | 38 | 56 | 5155  | 5793  | 0.095455343 | 0.404255319 | 0.595744681 | 0.904544657 |
| Reference 4 | 150 | 5  | 40 | 11461 | 11656 | 0.012918784 | 0.111111111 | 0.888888889 | 0.987081216 |
| Reference 5 | 64  | 3  | 10 | 2347  | 2424  | 0.026545002 | 0.230769231 | 0.769230769 | 0.973454998 |
| Reference 6 | 53  | 19 | 7  | 1207  | 1286  | 0.042063492 | 0.730769231 | 0.269230769 | 0.957936508 |

|                 |     |    |    |       |       |             |             |             |             |
|-----------------|-----|----|----|-------|-------|-------------|-------------|-------------|-------------|
| Reference<br>7  | 149 | 3  | 26 | 1720  | 1898  | 0.079721776 | 0.103448276 | 0.896551724 | 0.920278224 |
| Reference<br>8  | 119 | 4  | 10 | 3311  | 3444  | 0.034693878 | 0.285714286 | 0.714285714 | 0.965306122 |
| Reference<br>9  | 1   | 15 | 0  | 567   | 583   | 0.001760563 | 1           | 0           | 0.998239437 |
| Reference<br>10 | 59  | 2  | 11 | 2492  | 2564  | 0.023128185 | 0.153846154 | 0.846153846 | 0.976871815 |
| Reference<br>11 | 9   | 12 | 8  | 135   | 164   | 0.0625      | 0.6         | 0.4         | 0.9375      |
| Reference<br>12 | 200 | 12 | 32 | 7884  | 8128  | 0.024740228 | 0.272727273 | 0.727272727 | 0.975259772 |
| Reference<br>13 | 111 | 14 | 9  | 11804 | 11938 | 0.009315988 | 0.608695652 | 0.391304348 | 0.990684012 |

|                 |     |    |     |      |      |             |             |             |             |
|-----------------|-----|----|-----|------|------|-------------|-------------|-------------|-------------|
| Reference<br>14 | 7   | 11 | 2   | 308  | 328  | 0.022222222 | 0.846153846 | 0.153846154 | 0.977777778 |
| Reference<br>15 | 596 | 5  | 137 | 1090 | 1828 | 0.353499407 | 0.035211268 | 0.964788732 | 0.646500593 |

### Result Matrix:

|             | mean        | std         | min         | 25%         | 50%         | 75%         | max         | range       |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| FP Rate     | 0.057648281 | 0.086422851 | 0.001760563 | 0.017570503 | 0.026545002 | 0.064449039 | 0.353499407 | 0.351738843 |
| FN Rate     | 0.421272971 | 0.302487311 | 0.035211268 | 0.192307692 | 0.285714286 | 0.652173913 | 1           | 0.964788732 |
| Sensitivity | 0.578727029 | 0.302487311 | 0           | 0.347826087 | 0.714285714 | 0.807692308 | 0.964788732 | 0.964788732 |
| Specificity | 0.942351719 | 0.086422851 | 0.646500593 | 0.935550961 | 0.973454998 | 0.982429497 | 0.998239437 | 0.351738843 |
| NPV         | 0.987972601 | 0.021839228 | 0.918367347 | 0.988592452 | 0.998258851 | 0.998804365 | 0.999563928 | 0.081196581 |
| PPV         | 0.143664969 | 0.109100219 | 0           | 0.077648579 | 0.135135135 | 0.172022997 | 0.470588235 | 0.470588235 |

## References:

1. Gestational diabetes mellitus and development of intergenerational overall and subtypes of cardiovascular diseases: a systematic review and meta-analysis
2. Gestational diabetes mellitus and development of intergenerational non-alcoholic fatty liver disease (NAFLD) after delivery: a systematic review and meta-analysis
3. Comparative efficacy and acceptability of non-surgical brain stimulation for the acute treatment of major depressive episodes in adults: systematic review and network meta-analysis
4. Environmental toxic metal contaminants and risk of cardiovascular disease: systematic review and meta-analysis
5. Association Between Plant-Based Dietary Patterns and Risk of Type 2 Diabetes
6. The association of atherogenic index of plasma with cardiovascular outcomes in patients with coronary artery disease: A systematic review and meta-analysis
7. Genetic Polymorphisms and Clopidogrel Efficacy for Acute Ischemic Stroke or Transient Ischemic Attack: A Systematic Review and Meta-Analysis
8. Efficacy of Grass Pollen Allergen Sublingual Immunotherapy Tablets for Seasonal Allergic Rhinoconjunctivitis: A Systematic Review and Meta-analysis



9. Diagnostic Performance of Random Urine Samples Using Albumin Concentration vs Ratio of Albumin to Creatinine for Microalbuminuria Screening in Patients With Diabetes Mellitus
10. Effect of Anti-inflammatory Treatment on Depression, Depressive Symptoms, and Adverse Effects: A Systematic Review and Meta-analysis of Randomized Clinical Trials
11. Efficacy and safety of bempedoic acid for the treatment of hypercholesterolemia: A systematic review and meta-analysis
12. Association of puberty timing with type 2 diabetes: A systematic review and meta-analysis
13. Psychiatric Comorbidities in Persons With Epilepsy Compared With Persons Without Epilepsy
14. Quantification of glucose-6-phosphate dehydrogenase activity by spectrophotometry: A systematic review and meta-analysis
15. Preoperative/Neoadjuvant Therapy in Pancreatic Cancer: A Systematic Review and Meta-analysis of Response and Resection Percentages